

```

#include <iostream>
using namespace std;

class Student {
private:
    int id;
    float score;
public:
    Student(int i = 0, float s = 0.0f) : id(i), score(s) {}
    int getId() const { return id; }
    float getScore() const { return score; }
    void setInfo(int i, float s) {
        id = i;
        score = s;
    }

    void display() const {
        cout << "学号: " << id << "\t成绩: " << score << endl;
    }
};

// max函数
void max(Student *students, int n) {
    Student *maxStudent = students;
    float maxScore = students[0].getScore();

    for (int i = 1; i < n; i++) {
        if (students[i].getScore() > maxScore) {
            maxScore = students[i].getScore();
            maxStudent = &students[i];
        }
    }

    cout << "\n成绩最高的学生信息: " << endl;
    cout << "学号: " << maxStudent->getId()
    << "\t成绩: " << maxStudent->getScore() << endl;
}

int main() {
    Student students[5];
    students[0].setInfo(2301, 77.8);
    students[1].setInfo(2302, 81.0);
    students[2].setInfo(2303, 78.7);
    students[3].setInfo(2304, 67.2);
    students[4].setInfo(2305, 85.9);
    Student *ptr = students;

    cout << "第1个学生: ";
    ptr->display();
    ptr += 2;
    cout << "第3个学生: ";
    ptr->display();
    ptr += 2;
    cout << "第5个学生: ";
    ptr->display();
    max(students, 5);

    return 0;
}

```

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第1个学生: 学号: 2301    成绩: 77.8
第3个学生: 学号: 2303    成绩: 78.7
第5个学生: 学号: 2305    成绩: 85.9

```

```

成绩最高的学生信息:
学号: 2305    成绩: 85.9

```

```

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Process exited after 0.3756 seconds with
请按任意键继续. . . |

```

```

#include <iostream>
#include <cmath>
using namespace std;

class Complex {
private:
    double real;
    double imag;

public:
    Complex(double r = 0.0, double i = 0.0) : real(r), imag(i) {}

    Complex operator+(const Complex &c) const {
        return Complex(real + c.real, imag + c.imag);
    }

    Complex operator-(const Complex &c) const {
        return Complex(real - c.real, imag - c.imag);
    }

    Complex operator*(const Complex &c) const {
        return Complex(real * c.real - imag * c.imag,
            real * c.imag + imag * c.real);
    }

    Complex operator/(const Complex &c) const {
        double denominator = c.real * c.real + c.imag * c.imag;
        if (fabs(denominator) < 1e-10) {
            cout << "错误: 除数不能为零!" << endl;
            return Complex(0, 0);
        }
        return Complex((real * c.real + imag * c.imag) / denominator,
            (imag * c.real - real * c.imag) / denominator);
    }

    void display() const {
        if (imag >= 0)
            cout << real << " + " << imag << "i";
        else
            cout << real << " - " << -imag << "i";
    }

    void setComplex(double r, double i) {
        real = r;
        imag = i;
    }
};

```

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int main() {
    Complex c1(1.0, 3.0);
    Complex c2(4.0, -5.0);
    Complex result;

    cout << "复数 c1 = ";
    c1.display();
    cout << endl;

    cout << "复数 c2 = ";
    c2.display();
    cout << endl << endl;

    result = c1 + c2;
    cout << "加法结果: ";
    c1.display();
    cout << " + ";
    c2.display();
    cout << " = ";
    result.display();
    cout << endl;

    result = c1 - c2;
    cout << "减法结果: ";
    c1.display();
    cout << " - ";
    c2.display();
    cout << " = ";
    result.display();
    cout << endl;

    result = c1 * c2;
    cout << "乘法结果: ";
    c1.display();
    cout << " * ";
    c2.display();
    cout << " = ";
    result.display();
    cout << endl;

    result = c1 / c2;
    cout << "除法结果: ";
    c1.display();
    cout << " / ";
    c2.display();
    cout << " = ";
    result.display();
    cout << endl;

    return 0;
}

```

复数 c1 = 1 + 3i
 复数 c2 = 4 - 5i

加法结果: 1 + 3i + 4 - 5i = 5 - 2i

减法结果: 1 + 3i - 4 - 5i = -3 + 8i

乘法结果: 1 + 3i * 4 - 5i = 19 + 7i

除法结果: 1 + 3i / 4 - 5i = -0.268293 + 0.414634i

```

#include <iostream>
using namespace std;

class Matrix2x3 {
private:
    int data[2][3];
public:
    Matrix2x3() {
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 3; j++) {
                data[i][j] = 0;
            }
        }
    }

    Matrix2x3(int arr[2][3]) {
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 3; j++) {
                data[i][j] = arr[i][j];
            }
        }
    }

    friend istream& operator>>(istream& in, Matrix2x3& mat) {
        cout << "请输入2行3列矩阵的元素（按行输入，每行3个元素）：" << endl;
        for (int i = 0; i < 2; i++) {
            cout << "第" << i + 1 << "行: ";
            for (int j = 0; j < 3; j++) {
                in >> mat.data[i][j];
            }
        }
        return in;
    }
};

```

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friend ostream& operator<<(ostream& out, const Matrix2x3& mat) {
    out << "矩阵内容: " << endl;
    for (int i = 0; i < 2; i++) {
        out << "[";
        for (int j = 0; j < 3; j++) {
            out << mat.data[i][j];
            if (j < 2) out << ", ";
        }
        out << "]" << endl;
    }
    return out;
}

int getElement(int row, int col) const {
    if (row >= 0 && row < 2 && col >= 0 && col < 3) {
        return data[row][col];
    }
    return 0;
}

void setElement(int row, int col, int value) {
    if (row >= 0 && row < 2 && col >= 0 && col < 3) {
        data[row][col] = value;
    }
}

};

int main() {
    Matrix2x3 mat1, mat2;

    cout << "\n请输入第一个矩阵: " << endl;
    cin >> mat1;

    cout << "\n请输入第二个矩阵: " << endl;
    cin >> mat2;

    cout << "\n第一个矩阵: " << endl;
    cout << mat1;

    cout << "\n第二个矩阵: " << endl;
    cout << mat2;

    return 0;
}

```

请输入第一个矩阵:
 请输入2行3列矩阵的元素 (按行输入, 每行3个元素):
 第1行: 2 4 6
 第2行: 1 3 5

请输入第二个矩阵:
 请输入2行3列矩阵的元素 (按行输入, 每行3个元素):
 第1行: 12 34 45
 第2行: 67 78 89

第一个矩阵:
 矩阵内容:
 [2, 4, 6]
 [1, 3, 5]

第二个矩阵:
 矩阵内容:
 [12, 34, 45]
 [67, 78, 89]

```

#include <iostream>
#include <iomanip>
using namespace std;

class Date;

class Time {
private:
    int hour;
    int minute;
    int second;

public:
    Time(int h = 0, int m = 0, int s = 0) : hour(h), minute(m), second(s) {}

    void setTime(int h, int m, int s) {
        hour = h;
        minute = m;
        second = s;
    }

    void display(const Date &d);
};

class Date {
private:
    int year;
    int month;
    int day;

    friend class Time;

public:
    Date(int y = 2000, int m = 1, int d = 1) : year(y), month(m), day(d) {}

    void setDate(int y, int m, int d) {
        year = y;
        month = m;
        day = d;
    }

    void showDate() const {
        cout << year << "年" << month << "月" << day << "日";
    }
};

```

```

void Time::display(const Date &d) {
    cout << "当前日期和时间: " << endl;
    cout << "日期: ";
    d.showDate();
    cout << endl;

    cout << "时间: ";
    cout << setfill('0') << setw(2) << hour << ":"
        << setfill('0') << setw(2) << minute << ":"
        << setfill('0') << setw(2) << second << endl;

    cout << "详细信息: " << d.year << "年" << d.month << "月"
        << d.day << "日 "
        << setfill('0') << setw(2) << hour << "时"
        << setfill('0') << setw(2) << minute << "分"
        << setfill('0') << setw(2) << second << "秒" << endl;
}

int main() {
    Date today(2025, 11, 21);
    Time now(17, 12, 15);

    now.display(today);

    int y, m, d, h, min, s;

    cout << "请输入日期 (年 月 日): ";
    cin >> y >> m >> d;

    cout << "请输入时间 (时 分 秒): ";
    cin >> h >> min >> s;

    Date userDate(y, m, d);
    Time userTime(h, min, s);

    userTime.display(userDate);

    return 0;
}

```

```

当前日期和时间:
日期: 2025年11月21日
时间: 17:12:15
详细信息: 2025年11月21日 17时12分15秒
请输入日期 (年 月 日): 2025 8 1
请输入时间 (时 分 秒): 18 18 18
当前日期和时间:
日期: 2025年8月1日
时间: 18:18:18
详细信息: 2025年8月1日 18时18分18秒
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```